INCH-POUND
MIL-DTL-0016377/87(SH)
30 June 2014

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; LAMP, SOLID STATE, COMMERCIAL, FOR USE AS REPLACEMENT FOR COMMERCIAL F20T12 FLUORESCENT LAMP IN NON-VITAL LIGHT FIXTURES

This specification is approved for interim use by the Naval Sea Systems Command, Department of the Navy. Other activities in the Department of Defense may use this interim document.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

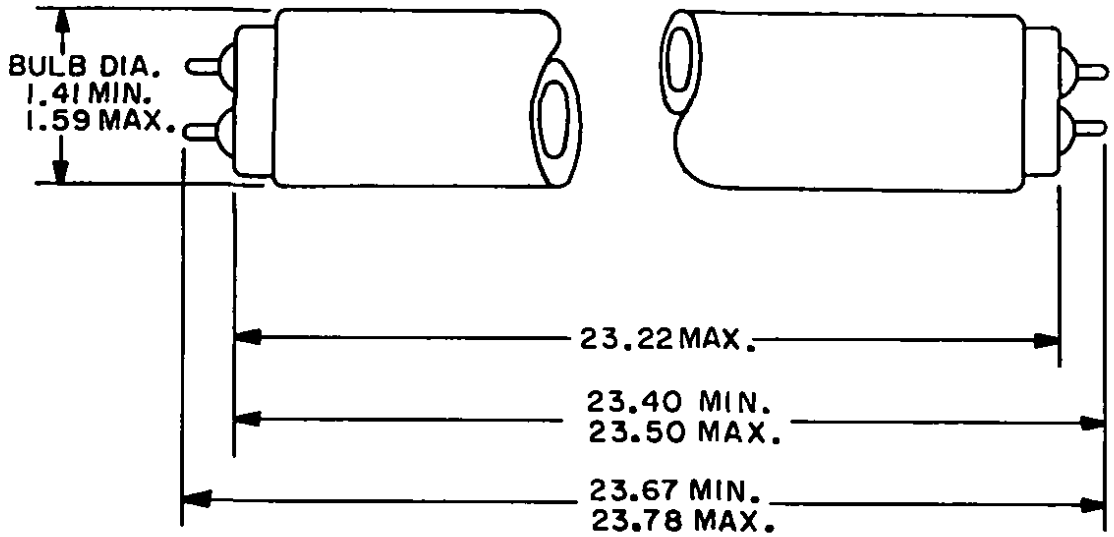


FIGURE 1. Lamp dimensions (inches).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:57Z
Check the source to verify that this is the current version before use.

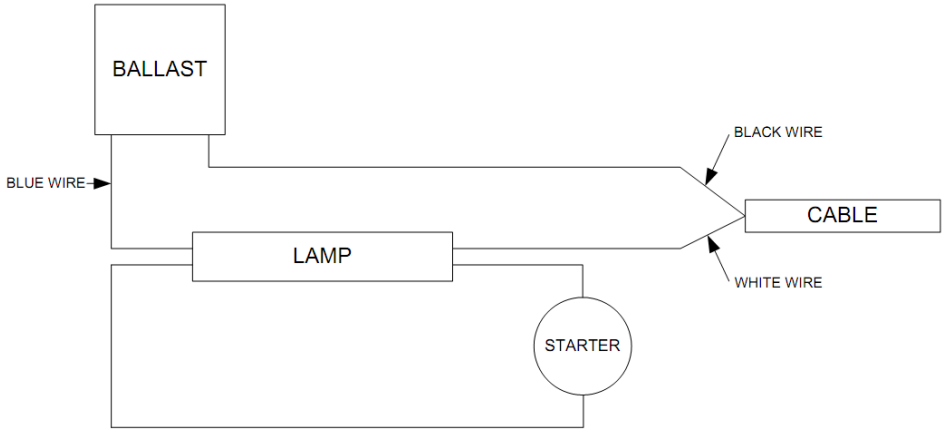


FIGURE 2. Wiring diagram for single fluorescent lamp utilizing M16377/44, 2-wire ballast.

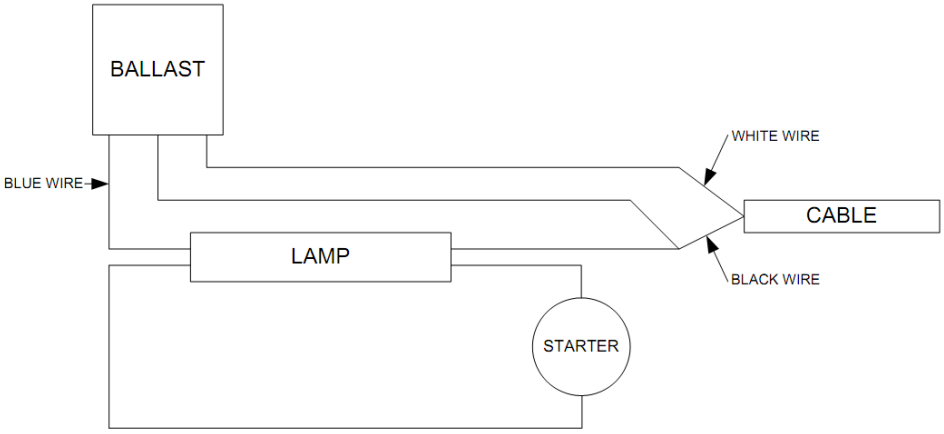


FIGURE 3. Wiring diagram for single fluorescent lamp utilizing M16377/77, 3-wire ballast.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:57Z
Check the source to verify that this is the current version before use.

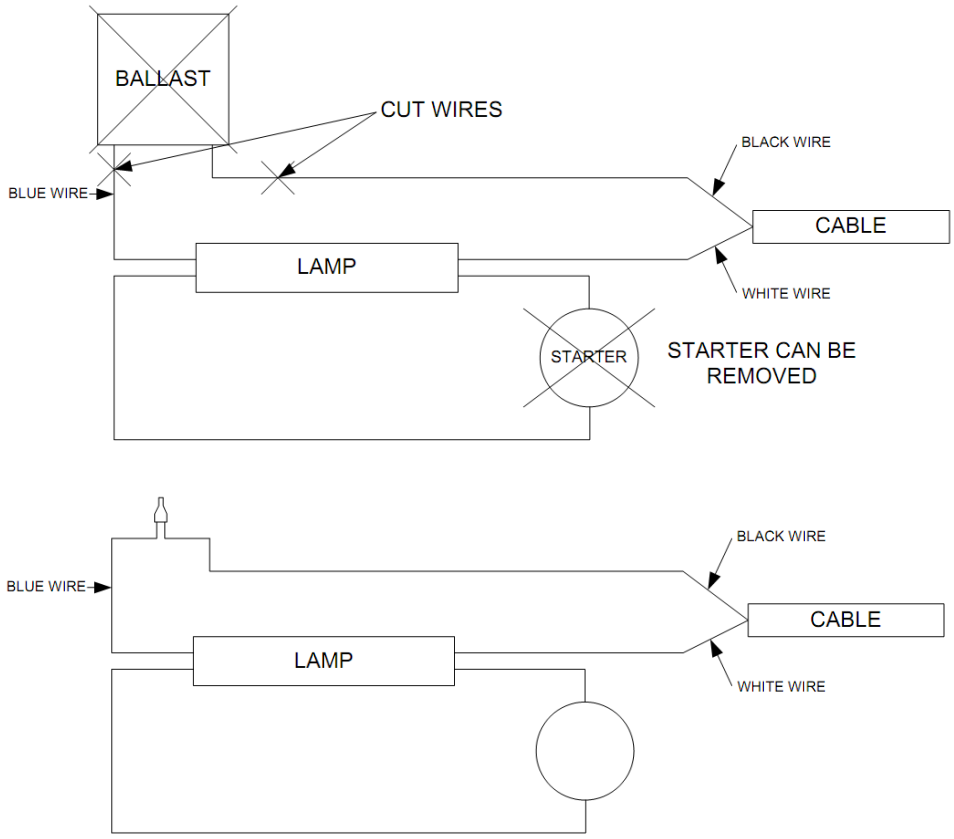


FIGURE 4. Ballast and starter removal for fixtures utilizing M16377/44, 2-wire ballasts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:57Z
Check the source to verify that this is the current version before use.

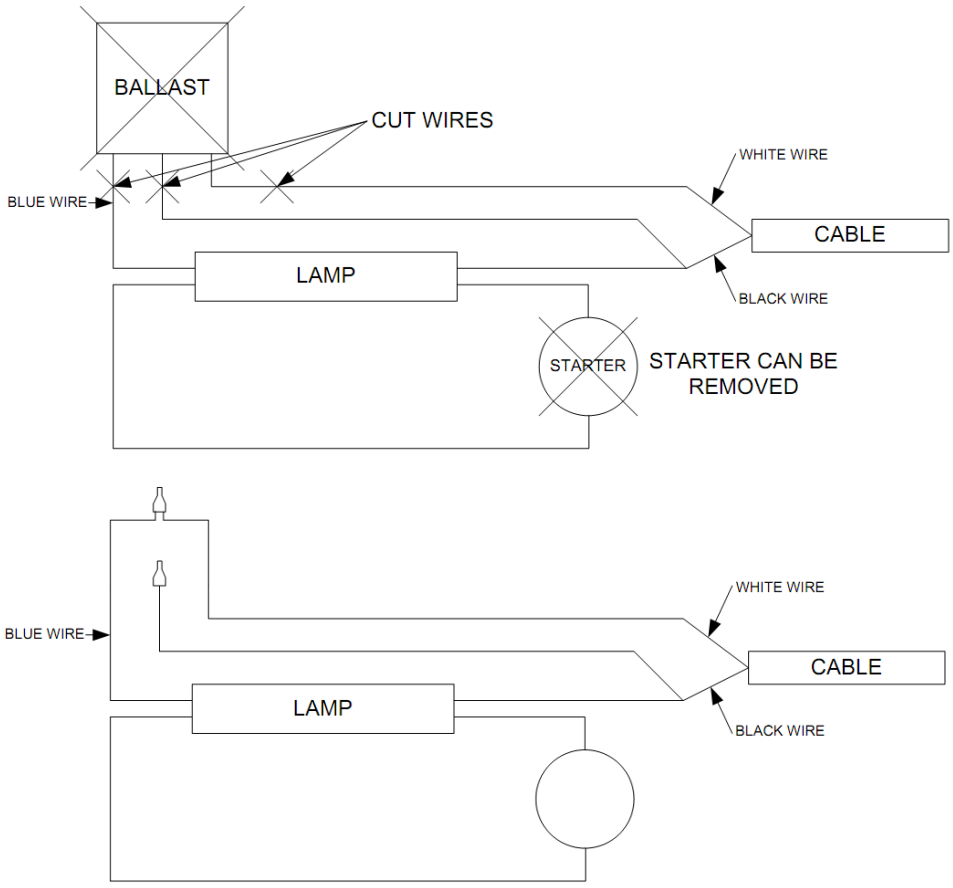


FIGURE 5. Ballast and starter removal for fixtures utilizing M16377/77, 3-wire ballast.

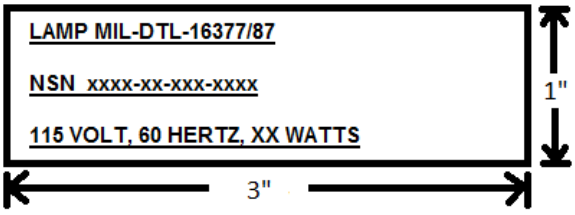


FIGURE 6. Label for lamp.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:57Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

DESIGN:

Safety:	Shall meet requirements of UL 1993.
Electrical rating:	12 watts, maximum.
Dimensions:	See figure 1.
Nominal bulb diameter:	1½ inches.
Nominal overall length:	24 inches.
Operation:	The solid state lamp shall utilize light emitting diodes (LEDs) and is intended for use as a replacement of the F20T12 and F17T12 fluorescent lamps used in the following MIL-DTL-16377 fixtures installed in non-vital spaces: M16377/8 M16377/11 M16377/12 M16377/57 Colored lighting shall accomplished utilizing existing M16377/42 filters.
Base designation:	Compatible with medium bipin base of figure SB12 of UL 496 (ANSI G13/G13M).
Weight:	0.95 pounds maximum.
Color rendering index (CRI):	≥75.
Correlated color temperature:	4000 K in accordance with ANSI ANSLG C78.377.
Rated life:	L70 ≥ 36,000 hours in accordance with IES TM-21.
Average initial lumens:	1050-1400.
Electrical operation:	Lamp shall be compatible with M16377 fixtures utilizing the following ballasts and with the starter removed: M16377/44 – figure 2 M16377/72 – figure 3 Lamp shall also function in M16377 fixtures when the starter has been removed and the ballast has been removed in accordance with figure 4 or figure 5 and ungrounded 115 volts alternating current (VAC) is applied to the lamp holders.
Circuit design:	The lamp assembly shall incorporate integrated driver circuitry so that no external components are required.
Designation and marking:	One label as shown on figure 6.

EXAMINATION AND TESTS:

First article inspection:	Examination and test shall be as specified in UL 1993 and as in table I.
Conformance:	
Conformance inspection:	See UL 1993.
Comparison inspection:	Same as the first article inspection.

TABLE I. First article and comparison inspection for complete T12 LED tube. 1/

Tests	Requirement	Test Method	Remarks
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT)	3.7.4.1.1	4.8.6.6	4000 K

NOTE:
1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:57Z
Check the source to verify that this is the current version before use.

Preparing activity:
Navy – SH
(Project 6210-2014-014)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:57Z
Check the source to verify that this is the current version before use.